AL-KAUSAR
COMPUTER INSTITUTE

KATAF ACADEMY

Course Syllabus 2026

9 hands-on courses, beginner and advanced, with weekly plans, estimated durations, projects and an internship for every course.

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01 / ABOUT THIS SYLLABUS

Built around doing the work.

Every course is a sequence of modules. Each module has topics, a hands-on lab you complete in class, and a deliverable you finish. Nothing is only watched.

SESSION

2 hours

Each class is a working session on your own computer.

WEEK

3 sessions

6 hours a week. Full Stack runs 4 sessions, 8 hours a week.

PLUS

Practice

Expect a few hours of your own practice each week. It is not counted in the totals.

All durations in this document are estimates for planning. Final batch length, dates and fees are confirmed in writing at admission.

Courses at a glance

BEGINNER NO EXPERIENCE NEEDED

COURSE	DURATION	SCHEDULE	TOTAL
Computer Essentials	10 weeks	3 × 2 h / week	**60 h**
Graphic Designing	14 weeks	3 × 2 h / week	**84 h**
Video Editing	12 weeks	3 × 2 h / week	**72 h**
Digital Marketing	12 weeks	3 × 2 h / week	**72 h**
AI Skills	9 weeks	3 × 2 h / week	**54 h**
Web Development	16 weeks	3 × 2 h / week	**96 h**

ADVANCED PROFESSIONAL TRACKS

COURSE	DURATION	SCHEDULE	TOTAL
Advanced Graphic Designing	20 weeks	3 × 2 h / week	**120 h**
Backend Development	16 weeks	3 × 2 h / week	**96 h**
Full Stack Development	32 weeks	4 × 2 h / week	**256 h**

02 / LEARNING PATHS

Where each path can take you.

Start at the level that matches you. Estimated time is the sum of the course durations.

PATH	COURSES, IN ORDER	ESTIMATED TIME
Office professional	Computer Essentials	**10 weeks**
Designer	Graphic Designing → Advanced Graphic Designing	**34 weeks**
Video editor	Video Editing	**12 weeks**
Digital marketer	Digital Marketing → AI Skills	**21 weeks**
Frontend developer	Web Development	**16 weeks**
Backend developer	Web Development → Backend Development	**32 weeks**
Full stack developer	Full Stack Development	**32 weeks**
Content creator	Graphic Designing → Video Editing → Digital Marketing	**38 weeks**

03 / HOW WE TEACH

We do not just teach. You get full hands-on experience.

01 **You work in every session**

Each class is a working session. You do the task yourself, on your own computer, with the instructor beside you.

02 **Every module ends in a build**

No module is finished until you have made something: a design, a page, an API, a video, a spreadsheet.

03 **Feedback on your own work**

We review what you made, show you what to fix, and you fix it.

04 **A capstone you can show**

Every course ends with a project for your portfolio.

How you are assessed, and what you receive.

ASSESSMENT

- ✓ A hands-on lab in every session, reviewed by the instructor.
- ✓ A deliverable at the end of every module.
- ✓ A capstone project reviewed against a clear checklist.

CERTIFICATE

- ✓ Awarded when every module deliverable and the capstone are complete.
- ✓ Carries your name, the course and a unique ID.
- ✓ Anyone can verify it online at kataf.pk/verify.

INTERNSHIP

An internship follows every course, working with the team on real tasks with feedback. Estimated length: **4 to 8 weeks**, part-time. Dates, hours and terms are explained at admission, before you pay.

BATCHES

Men and women study in separate batches with their own timings. Choose your batch and a preferred slot on the admission form.

COURSE 1 OF 9

BEGINNER

Computer Essentials

10	3 × 2 h	60 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

The computer skills every office job asks for. You start with using a computer and the keyboard, then build real documents, spreadsheets and presentations, and finish by working in a team on shared files.

WHO IT IS FOR

Students, job seekers, office staff and business owners. No computer experience needed.

BEFORE YOU START

None.

BY THE END YOU CAN

- ✓ Use Windows, files and folders confidently, and type at a useful speed.
- ✓ Write, format and print professional documents in Microsoft Word.
- ✓ Build spreadsheets with formulas, tables, charts and pivot tables in Excel.
- ✓ Design and deliver a clear presentation in PowerPoint.
- ✓ Work on shared files with Google Workspace and stay safe online.

TOOLS: Windows · Microsoft Word · Microsoft Excel · Microsoft PowerPoint · Google Workspace · Gmail / Outlook

Plan at a glance

MODULE	WHEN	HOURS
01 Computer and Windows basics	Week 1	6 h
02 Typing and keyboard shortcuts	Week 2	6 h
03 Internet, email and online safety	Week 3	6 h
04 Microsoft Word	Weeks 4–5	12 h
05 Microsoft Excel	Weeks 6–7	12 h
06 Microsoft PowerPoint	Week 8	6 h
07 Google Workspace	Week 9	6 h
08 Capstone project	Week 10	6 h
Total	**10 weeks**	**60 h**

Module by module

01 Computer and Windows basics

WEEK 1 · 6 H

TOPICS

- → Hardware and software, in plain words
- → The Windows desktop, settings and shortcuts
- → Files, folders, drives and USB storage
- → Copying, moving, searching and recovering files
- → Printers, scanners and everyday setup
- → Installing and removing programs safely

HANDS-ON LAB

Set up a clean folder system for a small office and print a test document.

YOU FINISH WITH

An organised file structure and a one-page setup checklist.

02 Typing and keyboard shortcuts

WEEK 2 · 6 H

TOPICS

- → Home-row touch typing and posture
- → Accuracy first, then speed
- → The 25 shortcuts that save the most time
- → Typing in English and Urdu
- → Daily timed drills with progress tracking

HANDS-ON LAB

Short timed typing drills every session, with a score recorded each time.

YOU FINISH WITH

A typing progress sheet showing speed and accuracy over the week.

03 Internet, email and online safety

WEEK 3 · 6 H

TOPICS

- → Browsers, tabs, bookmarks and downloads
- → Searching properly and judging sources
- → Creating and using a professional email account
- → Writing clear emails, attachments and replies
- → Passwords, two-step login and phishing
- → Backups and keeping your accounts safe

HANDS-ON LAB

Send a professional email with an attachment, and spot the fake in a set of phishing examples.

YOU FINISH WITH

A professional email account, a signature and a personal safety checklist.

04 Microsoft Word

WEEKS 4–5 · 12 H

TOPICS

- → Typing, selecting and formatting text
- → Paragraphs, spacing, lists and indents
- → Styles and keeping a document consistent
- → Page layout, margins, headers, footers and page numbers
- → Tables, images and shapes
- → Table of contents, mail merge and track changes
- → Saving as PDF and printing correctly

HANDS-ON LAB

Rebuild three real documents from messy drafts: a CV, a formal letter and a multi-page report.

YOU FINISH WITH

A formatted CV, a cover letter and a 5-page report with a table of contents.

05 Microsoft Excel

WEEKS 6–7 · 12 H

TOPICS

- → Cells, rows, columns and formatting
- → Formulas and functions: SUM, AVERAGE, IF, COUNTIF, SUMIF
- → XLOOKUP and VLOOKUP for finding data
- → Sorting, filtering and tables
- → Charts that explain the numbers
- → Pivot tables for quick summaries
- → Data validation and printing a clean report

HANDS-ON LAB

Build a monthly expense tracker, then a sales sheet with a summary chart.

YOU FINISH WITH

An expense tracker and a sales report with a pivot table and chart.

06 Microsoft PowerPoint

WEEK 8 · 6 H

TOPICS

- → Slide layouts and the slide master
- → Design rules: one idea per slide
- → Images, icons, charts and tables
- → Animation and transitions, used sparingly
- → Presenting: pace, voice and notes

HANDS-ON LAB

Turn a long document into a ten-slide presentation and present it to the class.

YOU FINISH WITH

A ten-slide presentation, delivered live.

07 Google Workspace

WEEK 9 · 6 H

TOPICS

- → Google Docs, Sheets and Slides
- → Sharing, comments and version history
- → Drive, folders and permissions
- → Google Forms for surveys and sign-ups
- → Working with Microsoft files in Google

HANDS-ON LAB

Work in pairs on one shared document, sheet and form at the same time.

YOU FINISH WITH

A shared team project: a document, a sheet and a form.

08 Capstone project

WEEK 10 · 6 H

TOPICS

- → Choose a real small-business scenario
- → Plan the documents, sheet and presentation you need
- → Build, review and revise
- → Present to the class

HANDS-ON LAB

A full office pack for one small business, built from scratch.

YOU FINISH WITH

A complete office pack: letterhead, price sheet with formulas, and a business presentation.

CAPSTONE PROJECT

A complete office pack for a small business: a letter and CV set in Word, a price and sales sheet with formulas and a chart in Excel, and a ten-slide presentation.

WHERE IT LEADS

- ✓ Office assistant
- ✓ Data entry operator
- ✓ Administrator
- ✓ Receptionist
- ✓ Business owner who runs their own paperwork

INTERNSHIP ROLE

Office and data intern. Work alongside the team on real tasks, with feedback.

GOOD NEXT STEPS

Digital Marketing | AI Skills | Web Development

COURSE 2 OF 9

BEGINNER

Graphic Designing

14	3 × 2 h	84 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

A first full course in design. You learn how to see layout, colour and type, then build everyday work in Photoshop, Illustrator and Canva: logos, posters, visiting cards, social media posts and a starter portfolio.

WHO IT IS FOR

School and college students, press and shop staff moving into design, and anyone starting freelance design work. You do not need to draw.

BEFORE YOU START

None. You start from zero.

BY THE END YOU CAN

- ✓ Apply layout, hierarchy, colour and typography to any design.
- ✓ Edit and compose images in Photoshop.
- ✓ Draw logos and icons as vectors in Illustrator.
- ✓ Prepare print-ready files for a real press.
- ✓ Present a starter portfolio of three to five finished pieces.

TOOLS: Adobe Photoshop | Adobe Illustrator | Canva | A pen and sketchbook

Plan at a glance

MODULE	WHEN	HOURS
01 Design foundations	Weeks 1–2	12 h
02 Colour and typography	Weeks 3–4	12 h
03 Photoshop essentials	Weeks 5–7	18 h
04 Illustrator essentials	Weeks 8–10	18 h
05 Canva and quick design	Week 11	6 h
06 Print basics	Week 12	6 h
07 Portfolio and capstone	Weeks 13–14	12 h
Total	**14 weeks**	**84 h**

Module by module

01 Design foundations

WEEKS 1-2 · 12 H

TOPICS

- → What a client brief is, and how to read one
- → Layout, grids and alignment
- → Visual hierarchy and focus
- → Space, balance and contrast
- → Gestalt: how people group what they see
- → Looking at good and bad design critically

HANDS-ON LAB

Redesign three poorly made flyers using only layout and hierarchy, no new images.

YOU FINISH WITH

Three before-and-after flyer redesigns with a one-line reason for each change.

02 Colour and typography

WEEKS 3-4 · 12 H

TOPICS

- → Colour theory: hue, saturation, brightness
- → Building a palette and checking contrast
- → Typeface families, weights and pairing
- → Readable text: size, spacing and line length
- → English and Urdu typography
- → Putting type on images without losing it

HANDS-ON LAB

Build two palettes and two font pairings, then apply each to the same poster.

YOU FINISH WITH

A two-poster set with documented palette and type choices.

03 Photoshop essentials

WEEKS 5-7 · 18 H

TOPICS

- → The interface, documents and resolution
- → Layers, masks and smart objects
- → Selections and cut-outs
- → Retouching: spot healing, colour and exposure correction
- → Text, shapes and effects
- → Social media post templates
- → Exporting for web and print

HANDS-ON LAB

Retouch a set of product photos, then compose a social media post from them.

YOU FINISH WITH

A retouched photo set and three social media posts.

04 Illustrator essentials

WEEKS 8-10 · 18 H

TOPICS

- → Vector vs raster, and why it matters
- → The pen tool and shape builder
- → Paths, anchor points and curves
- → Colour, gradients and strokes
- → Building a simple logo from sketches
- → Icons and simple illustration
- → Artboards and exporting SVG, PDF and PNG

HANDS-ON LAB

Sketch, then draw a logo for a local business and test it at small and large sizes.

YOU FINISH WITH

A logo with colour, black and white, and small-size versions.

05 Canva and quick design

WEEK 11 · 6 H

TOPICS

- → Using templates properly, not copying them
- → Brand colours and fonts in Canva
- → Posts, banners and short presentations
- → Working fast without losing quality

HANDS-ON LAB

Produce a week of social content for one shop in a single sitting.

YOU FINISH WITH

A seven-post content set in one consistent style.

06 Print basics

WEEK 12 · 6 H

TOPICS

- → Bleed, trim and safe margins
- → CMYK vs RGB
- → Resolution and file formats
- → Visiting cards, flyers and flex banners
- → Talking to a press and checking a proof

HANDS-ON LAB

Prepare a visiting card and flyer, and review a printed proof.

YOU FINISH WITH

A print-ready visiting card and flyer.

07 Portfolio and capstone

WEEKS 13–14 · 12 H

TOPICS

- → Choosing your strongest work
- → Writing a short case study for each
- → Presenting work to a client
- → Setting up a portfolio on Behance or a simple site
- → Final project review

HANDS-ON LAB

Design a complete mini brand: logo, card, flyer and three social posts.

YOU FINISH WITH

A starter portfolio of three to five pieces, including the mini brand.

CAPSTONE PROJECT

A mini brand for a real local business: logo, visiting card, flyer and a three-post social media set, presented as a short case study.

WHERE IT LEADS

- ✓ Junior graphic designer
- ✓ Social media designer
- ✓ Print designer
- ✓ Freelance designer

INTERNSHIP ROLE

Design intern. Work alongside the team on real tasks, with feedback.

GOOD NEXT STEPS

Advanced Graphic Designing | Video Editing | Digital Marketing

COURSE 3 OF 9

BEGINNER

Video Editing

12	3 × 2 h	72 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

From raw footage to finished videos people want to watch. You learn the editing workflow in Premiere Pro and DaVinci Resolve, then practise on the formats clients actually pay for: short-form social video, product promos and event highlights.

WHO IT IS FOR

Students, content creators, shop owners and anyone starting freelance editing. No experience needed.

BEFORE YOU START

None. A mid-range laptop is enough to start.

BY THE END YOU CAN

- ✓ Edit a video from raw footage to a finished export.
- ✓ Clean up and mix audio, music and voiceover.
- ✓ Correct and grade colour so shots match.
- ✓ Add titles, captions and simple motion graphics.
- ✓ Deliver files in the right format for each platform.

TOOLS Adobe Premiere Pro | DaVinci Resolve | CapCut | Adobe After Effects (basics)

Plan at a glance

MODULE	WHEN	HOURS
01 Video fundamentals and workflow	Week 1	6 h
02 Editing with Premiere Pro	Weeks 2–3	12 h
03 Audio	Week 4	6 h
04 Colour with DaVinci Resolve	Weeks 5–6	12 h
05 Titles, captions and motion	Weeks 7–8	12 h
06 Short-form video	Week 9	6 h
07 Business videos	Week 10	6 h
08 Delivery and freelancing	Week 11	6 h
09 Capstone project	Week 12	6 h
Total	**12 weeks**	**72 h**

Module by module

01 Video fundamentals and workflow

WEEK 1 · 6 H

TOPICS

- → Frame rate, resolution and aspect ratio
- → Codecs and formats, in plain words
- → Organising footage and project files
- → Backing up and naming files
- → How an edit is planned from brief to delivery

HANDS-ON LAB

Set up and organise a project from a folder of mixed raw clips.

YOU FINISH WITH

A tidy project with bins, naming and a backup.

02 Editing with Premiere Pro

WEEKS 2-3 · 12 H

TOPICS

- → The interface, timeline and tools
- → Cutting, trimming and ripple edits
- → J-cuts, L-cuts and cutting on action
- → Pacing, rhythm and story
- → Transitions, used with purpose
- → Speed changes and basic effects

HANDS-ON LAB

Cut the same interview twice: once slow, once tight, and compare.

YOU FINISH WITH

A 60-second edited piece with two versions.

03 Audio

WEEK 4 · 6 H

TOPICS

- → Why audio matters more than picture
- → Cleaning voice: noise, hum and levels
- → Adding music and sound effects
- → Mixing: ducking and balancing
- → Recording a clear voiceover

HANDS-ON LAB

Fix a badly recorded interview and add a music bed.

YOU FINISH WITH

A cleaned, mixed audio track for a short video.

04 Colour with DaVinci Resolve

WEEKS 5–6 · 12 H

TOPICS

- → Reading scopes: waveform and histogram
- → Exposure and white balance correction
- → Primary grading: lift, gamma, gain
- → Matching shots from different cameras
- → Looks and LUTs, and when not to use them
- → Skin tones and secondary corrections

HANDS-ON LAB

Correct and grade a mismatched set of clips into one consistent look.

YOU FINISH WITH

A graded sequence with before-and-after comparison.

05 Titles, captions and motion

WEEKS 7–8 · 12 H

TOPICS

- → Titles and lower thirds
- → Captions and subtitles
- → Keyframes and simple animation
- → Masks and basic compositing
- → Intro to After Effects
- → Templates and keeping a style consistent

HANDS-ON LAB

Build a branded title pack and caption a talking-head video.

YOU FINISH WITH

A reusable title pack and a captioned video.

06 Short-form video

WEEK 9 · 6 H

TOPICS

- → Reels, TikTok and Shorts: what each rewards
- → Vertical framing and safe zones
- → Hooks and holding attention
- → Fast cuts, captions and sound trends
- → Batch-editing a week of content

HANDS-ON LAB

Edit five vertical videos for one brand in a single session.

YOU FINISH WITH

A five-video short-form set.

07 Business videos

WEEK 10 · 6 H

TOPICS

- → Product videos and promos
- → Ads with a clear offer and call to action
- → Event, wedding and highlight reels
- → Working from a client brief and shot list

HANDS-ON LAB

Edit a product promo from a client brief with supplied footage.

YOU FINISH WITH

A 30-second product promo and a highlight reel.

08 Delivery and freelancing

WEEK 11 · 6 H

TOPICS

- → Export settings by platform
- → File size and quality trade-offs
- → Review rounds and client feedback
- → Quoting, proposals and pricing
- → Building a showreel

HANDS-ON LAB

Export one video for four platforms and write a proposal for a client.

YOU FINISH WITH

Platform exports and a one-page proposal.

09 Capstone project

WEEK 12 · 6 H

TOPICS

- → Plan, shoot or collect footage
- → Edit, grade, sound and title
- → Export and present

HANDS-ON LAB

Produce a finished video for a real or realistic client.

YOU FINISH WITH

A finished client-style video and a 60-second showreel.

CAPSTONE PROJECT

A complete promotional video for a local business, with audio, colour, titles and captions, plus a 60-second showreel of your best work.

WHERE IT LEADS

- ✓ Video editor
- ✓ Social media video editor
- ✓ Content creator
- ✓ Freelance editor

INTERNSHIP ROLE

Video editing intern. Work alongside the team on real tasks, with feedback.

GOOD NEXT STEPS

Digital Marketing | Advanced Graphic Designing

COURSE 4 OF 9

BEGINNER

Digital Marketing

12	3 × 2 h	72 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

How to get customers online and know what is working. You plan content, write offers, run paid campaigns on Meta and Google, use WhatsApp properly, and report results from real numbers. Ends with a live campaign.

WHO IT IS FOR

Shop and business owners, students entering marketing, and freelancers who want to run campaigns for clients.

BEFORE YOU START

None. Basic computer skills help.

BY THE END YOU CAN

- ✓ Plan a campaign around an audience, an offer and a budget.
- ✓ Create and schedule content for Facebook, Instagram and TikTok.
- ✓ Set up, run and optimise a Meta Ads campaign.
- ✓ Set up a Google Business Profile and understand search and SEO basics.
- ✓ Read analytics and report results to a client.

TOOLS Meta Business Suite and Ads Manager · Google Business Profile · Google Analytics · Canva · WhatsApp Business · AI writing tools

Plan at a glance

MODULE	WHEN	HOURS
01 Marketing foundations	Week 1	6 h
02 Social media strategy and content	Weeks 2–3	12 h
03 Copywriting and content	Week 4	6 h
04 Meta Ads	Weeks 5–6	12 h
05 Google: search, maps and SEO	Weeks 7–8	12 h
06 WhatsApp marketing	Week 9	6 h
07 Analytics and reporting	Week 10	6 h
08 Freelancing and agency work	Week 11	6 h
09 Capstone: a live campaign	Week 12	6 h
Total	**12 weeks**	**72 h**

Module by module

01 Marketing foundations

WEEK 1 · 6 H

TOPICS

- → Audience, offer and positioning
- → How customers decide to buy
- → Setting a goal and a budget
- → The funnel: attention, interest, action
- → Researching competitors

HANDS-ON LAB

Write a one-page marketing plan for a real local business.

YOU FINISH WITH

A one-page marketing plan.

02 Social media strategy and content

WEEKS 2-3 · 12 H

TOPICS

- → Choosing platforms for a business
- → Content pillars and a monthly calendar
- → Posts, carousels and short video
- → Captions that get replies
- → Community management
- → Scheduling with Meta Business Suite

HANDS-ON LAB

Build and schedule a 30-day content calendar for one business.

YOU FINISH WITH

A 30-day content calendar with the first week produced.

03 Copywriting and content

WEEK 4 · 6 H

TOPICS

- → Headlines, hooks and calls to action
- → Writing offers people understand
- → Product and service pages
- → Using AI to draft, then editing properly
- → Keeping one brand voice

HANDS-ON LAB

Rewrite weak product descriptions and ads, then test two versions.

YOU FINISH WITH

A set of improved ads and product descriptions.

04 Meta Ads

WEEKS 5-6 · 12 H

TOPICS

- → Ad account, pixel and business setup
- → Campaign, ad set and ad structure
- → Objectives and audiences
- → Budgets, bidding and learning phase
- → Creative testing: images, video and copy
- → Reading results: reach, clicks, cost per result
- → Retargeting

HANDS-ON LAB

Launch a small live campaign with a real budget and track it daily.

YOU FINISH WITH

A live campaign with a results log.

05 Google: search, maps and SEO

WEEKS 7-8 · 12 H

TOPICS

- → How search works and what people search for
- → Keyword research for local businesses
- → On-page SEO basics
- → Google Business Profile and reviews
- → Intro to Google Search Ads
- → Local search ranking factors

HANDS-ON LAB

Set up a complete Google Business Profile and a keyword plan for a business.

YOU FINISH WITH

A finished Business Profile and a keyword plan.

06 WhatsApp marketing

WEEK 9 · 6 H

TOPICS

- → WhatsApp Business app, catalogue and labels
- → Quick replies and greeting messages
- → Broadcast lists, used respectfully
- → Click-to-WhatsApp ads
- → Following up on enquiries

HANDS-ON LAB

Set up a WhatsApp Business account with catalogue and quick replies.

YOU FINISH WITH

A configured WhatsApp Business setup and a follow-up script.

07 Analytics and reporting

WEEK 10 · 6 H

TOPICS

- → What to measure, and what to ignore
- → Google Analytics basics
- → Meta insights
- → Building a simple dashboard
- → Writing a one-page client report

HANDS-ON LAB

Turn campaign numbers into a report a client can act on.

YOU FINISH WITH

A one-page performance report.

08 Freelancing and agency work

WEEK 11 · 6 H

TOPICS

- → Finding your first clients
- → Proposals, pricing and contracts
- → Managing several accounts
- → Setting expectations and reporting rhythm

HANDS-ON LAB

Write a proposal and a pricing sheet for a marketing package.

YOU FINISH WITH

A proposal and package pricing.

09 Capstone: a live campaign

WEEK 12 · 6 H

TOPICS

- → Plan, build and launch
- → Optimise during the run
- → Report results

HANDS-ON LAB

Run a campaign from plan to report for a real or realistic business.

YOU FINISH WITH

A campaign plan, creative, live results and a final report.

CAPSTONE PROJECT

A complete campaign for a real business: plan, content, a live Meta Ads test, Google Business Profile setup, WhatsApp setup and a results report.

WHERE IT LEADS

- ✓ Digital marketer
- ✓ Social media manager
- ✓ Performance marketing assistant
- ✓ Freelance marketer

INTERNSHIP ROLE

Marketing intern. Work alongside the team on real tasks, with feedback.

GOOD NEXT STEPS

AI Skills | Graphic Designing | Video Editing

COURSE 5 OF 9

BEGINNER

AI Skills

9	3 × 2 h	54 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

Use AI properly in real work. You learn how language models behave and why they make mistakes, write prompts that give reliable results, use AI tools for everyday tasks, automate repeated work, and build a small AI assistant.

WHO IT IS FOR

Students, freelancers, office staff and business owners. The tools and prompting modules need no coding.

BEFORE YOU START

None for the first five weeks. The API module is easier with basic programming.

BY THE END YOU CAN

- ✓ Explain what AI can and cannot do, and why it makes mistakes.
- ✓ Write clear prompts and check the results.
- ✓ Use AI for writing, research, design, documents and coding help.
- ✓ Build an automation that removes a repeated task.
- ✓ Build and test a small AI assistant for a business.

TOOLS: ChatGPT · Claude · Gemini · AI image and video tools · No-code automation tools · AI APIs

Plan at a glance

MODULE	WHEN	HOURS
01 How AI models work	Week 1	6 h
02 Prompting that works	Week 2	6 h
03 AI tools for work	Weeks 3–4	12 h
04 Automation	Week 5	6 h
05 Building with AI APIs	Weeks 6–7	12 h
06 An assistant for a business	Week 8	6 h
07 Capstone and responsible use	Week 9	6 h
Total	**9 weeks**	**54 h**

Module by module

01 How AI models work

WEEK 1 · 6 H

TOPICS

- → What a language model is, in plain words
- → Training, tokens and context
- → What it does well and badly
- → Why it invents facts, and how to catch it
- → Privacy: what not to paste into AI

HANDS-ON LAB

Run the same questions on three models and record where each is wrong.

YOU FINISH WITH

A comparison sheet of strengths and failures.

02 Prompting that works

WEEK 2 · 6 H

TOPICS

- → Giving clear instructions
- → Context, examples and format
- → Roles, steps and constraints
- → Checking, correcting and iterating
- → Building a prompt library

HANDS-ON LAB

Take ten real tasks from your own work and write a tested prompt for each.

YOU FINISH WITH

A personal prompt library of ten tested prompts.

03 AI tools for work

WEEKS 3–4 · 12 H

TOPICS

- → Writing, editing and summarising
- → Research with sources you can check
- → Documents, slides and spreadsheets with AI
- → Images and design with AI
- → Meeting notes, translation and voice
- → Coding help for non-programmers

HANDS-ON LAB

Complete one week of your normal work using AI where it helps, and note the time saved.

YOU FINISH WITH

A before-and-after log of tasks with time saved and quality checks.

04 Automation

WEEK 5 · 6 H

TOPICS

- → Triggers and actions
- → Connecting forms, sheets, email and chat
- → Repeating tasks automatically
- → Handling errors and checking results

HANDS-ON LAB

Build a workflow that takes form entries, summarises them with AI and sends a daily digest.

YOU FINISH WITH

A working automation.

05 Building with AI APIs

WEEKS 6–7 · 12 H

TOPICS

- → Calling a model from code
- → Messages, system prompts and settings
- → Handling responses and errors
- → Structured output
- → Keeping costs under control
- → Using your own documents as context

HANDS-ON LAB

Write a small script that answers questions from a set of your own documents.

YOU FINISH WITH

A small script that answers from supplied documents.

06 An assistant for a business

WEEK 8 · 6 H

TOPICS

- → Choosing a real use case
- → Writing the instructions and limits
- → Testing with real questions
- → Measuring where it fails
- → Hand-over: how staff use it

HANDS-ON LAB

Build a customer-question assistant for one business.

YOU FINISH WITH

A tested assistant with a failure report.

07 Capstone and responsible use

WEEK 9 · 6 H

TOPICS

- → Privacy of customer data
- → When not to trust AI
- → Telling clients honestly how AI was used
- → Final presentation

HANDS-ON LAB

Present your assistant and your automation, and explain their limits.

YOU FINISH WITH

A final demo, a limits statement and a short write-up.

CAPSTONE PROJECT

A small AI assistant and one automation for a real business use case, with a written note on its limits and how it should be used responsibly.

WHERE IT LEADS

- ✓ AI-skilled office professional
- ✓ Automation assistant
- ✓ Content and research assistant
- ✓ Freelancer offering AI-assisted services

INTERNSHIP ROLE

AI intern. Work alongside the team on real tasks, with feedback.

GOOD NEXT STEPS

Web Development | Digital Marketing | Backend Development

COURSE 6 OF 9

BEGINNER

Web Development

16	3 × 2 h	96 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

A complete frontend course. You learn how the web works, then build with HTML, CSS and JavaScript, use Git, style with Tailwind, make sites fast and accessible, and publish a real website for a real business.

WHO IT IS FOR

Beginners who want to become frontend developers, students looking for a first tech skill, and owners who want to understand how their website is made.

BEFORE YOU START

None. You should be comfortable using a computer.

BY THE END YOU CAN

- ✓ Build a responsive website from a design, without a template.
- ✓ Write JavaScript that makes a page interactive and loads data from an API.
- ✓ Use Git and GitHub to save and share code.
- ✓ Make pages accessible, fast and usable on small phones.
- ✓ Deploy a site on a real domain.

TOOLS HTML · CSS · JavaScript · Tailwind CSS · Git and GitHub · VS Code · Chrome DevTools

Plan at a glance

MODULE	WHEN	HOURS
01 How the web works and your setup	Week 1	6 h
02 HTML	Weeks 2–3	12 h
03 CSS	Weeks 4–6	18 h
04 JavaScript fundamentals	Weeks 7–9	18 h
05 DOM, events and APIs	Weeks 10–11	12 h
06 Git and GitHub	Week 12	6 h
07 Tailwind CSS	Week 13	6 h
08 Accessibility, performance and deployment	Week 14	6 h
09 Capstone project	Weeks 15–16	12 h
Total	**16 weeks**	**96 h**

Module by module

01 How the web works and your setup

WEEK 1 · 6 H

TOPICS

- → Browsers, servers, domains and DNS
- → What happens when a page loads
- → Setting up VS Code, Git and the terminal
- → Reading a design and a brief
- → How developers work day to day

HANDS-ON LAB

Set up your tools and publish a first one-page site.

YOU FINISH WITH

A working development setup and a first page online.

02 HTML

WEEKS 2-3 · 12 H

TOPICS

- → Document structure and semantic tags
- → Headings, text, lists, links and images
- → Forms and inputs
- → Tables, audio and video
- → Accessibility basics: alt text and labels
- → Writing HTML that validates

HANDS-ON LAB

Build a complete multi-page site structure with no styling.

YOU FINISH WITH

A semantic multi-page HTML site.

03 CSS

WEEKS 4-6 · 18 H

TOPICS

- → Selectors, specificity and the cascade
- → The box model and layout
- → Flexbox and Grid
- → Responsive design and media queries
- → Typography, colour and spacing systems
- → Transitions and simple animation
- → Starting mobile-first

HANDS-ON LAB

Style the HTML site, then rebuild a real landing page from a design file.

YOU FINISH WITH

A responsive landing page that matches a design.

04 JavaScript fundamentals

WEEKS 7-9 · 18 H

TOPICS

- → Variables, types and operators
- → Conditions and loops
- → Functions and scope
- → Arrays and objects
- → Array methods: map, filter, reduce
- → ES modules
- → Debugging in the browser

HANDS-ON LAB

Solve a set of small problems, then build a tip calculator and a to-do list.

YOU FINISH WITH

A set of solved exercises and two small apps.

05 DOM, events and APIs

WEEKS 10-11 · 12 H

TOPICS

- → Selecting and changing elements
- → Events and forms
- → Local storage
- → Fetching data with fetch and promises
- → Async and await
- → Handling loading and error states

HANDS-ON LAB

Build an app that loads real data from a public API and filters it.

YOU FINISH WITH

A small data-driven app.

06 Git and GitHub

WEEK 12 · 6 H

TOPICS

- → Commits, branches and merging
- → Working with a remote repository
- → Pull requests and reviewing code
- → Resolving conflicts
- → Writing a good README

HANDS-ON LAB

Collaborate on a repository with a partner using branches and pull requests.

YOU FINISH WITH

A repository with a clean history and a README.

07 Tailwind CSS

WEEK 13 · 6 H

TOPICS

- → Utility-first styling
- → Responsive and state variants
- → Building reusable components
- → Design tokens and themes

HANDS-ON LAB

Rebuild your landing page in Tailwind.

YOU FINISH WITH

The same page rebuilt in Tailwind.

08 Accessibility, performance and deployment

WEEK 14 · 6 H

TOPICS

- → Keyboard use and screen-reader basics
- → Colour contrast
- → Image optimisation and loading
- → Lighthouse and Core Web Vitals
- → Hosting, domains and HTTPS
- → Basic SEO

HANDS-ON LAB

Audit your site with Lighthouse, fix the issues and publish it.

YOU FINISH WITH

A published site with an audit report.

09 Capstone project

WEEKS 15–16 · 12 H

TOPICS

- → Gather a real brief from a local business
- → Plan pages and content
- → Design, build and test
- → Present, publish and hand over

HANDS-ON LAB

Build and launch a complete website for a real business.

YOU FINISH WITH

A live website for a real business and a project presentation.

CAPSTONE PROJECT

A complete, responsive, accessible website for a real local business, published on a live domain.

WHERE IT LEADS

- ✓ Frontend developer (junior)
- ✓ Website developer
- ✓ Freelance web developer

INTERNSHIP ROLE

Frontend intern. Work alongside the team on real tasks, with feedback.

GOOD NEXT STEPS

Backend Development | Full Stack Development | Advanced Graphic Designing

COURSE 7 OF 9

ADVANCED

Advanced Graphic Designing

20	3 × 2 h	120 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

For designers who already work in Photoshop and Illustrator and want to work at a professional level: complete brand identities, packaging, editorial layouts and campaigns, delivered the way agencies deliver them.

WHO IT IS FOR

Designers who finished our beginner course or can already work comfortably in Photoshop and Illustrator.

BEFORE YOU START

Comfortable with layers, masks and the pen tool. A portfolio of at least three pieces helps.

BY THE END YOU CAN

- ✓ Develop a complete brand identity from research to guidelines.
- ✓ Compose and retouch at a professional level.
- ✓ Lay out multi-page documents in InDesign.
- ✓ Design packaging from a dieline, ready for production.
- ✓ Run a design project from brief to delivery and price it.

TOOLS Adobe Photoshop · Adobe Illustrator · Adobe InDesign · Figma

Plan at a glance

MODULE	WHEN	HOURS
01 Brand strategy and identity	Weeks 1–3	18 h
02 Advanced Illustrator	Weeks 4–6	18 h
03 Advanced Photoshop	Weeks 7–9	18 h
04 Layout with InDesign	Weeks 10–12	18 h
05 Packaging design	Weeks 13–14	12 h
06 Visual design in Figma	Weeks 15–16	12 h
07 Print production and prepress	Week 17	6 h
08 Client workflow and pricing	Week 18	6 h
09 Capstone brand project	Weeks 19–20	12 h
Total	**20 weeks**	**120 h**

Module by module

01 Brand strategy and identity

WEEKS 1–3 · 18 H

TOPICS

- → Brand research: audience, competitors and positioning
- → Brand personality and voice
- → Naming and taglines
- → Moodboards and design direction
- → Logo systems: primary, secondary, icon
- → Brand guidelines: rules, spacing and misuse
- → Presenting a brand to a client

HANDS-ON LAB

Develop a brand identity from a written brief: research, direction, logo system and guideline pages.

YOU FINISH WITH

A brand identity document with a logo system and guidelines.

02 Advanced Illustrator

WEEKS 4–6 · 18 H

TOPICS

- → Custom lettering and logotypes
- → Complex vector illustration
- → Pattern and texture design
- → Icon sets on a grid
- → Isometric and infographic illustration
- → Working with large files and symbols

HANDS-ON LAB

Design a custom wordmark and a 24-icon set for the brand.

YOU FINISH WITH

A custom wordmark, an icon set and a pattern.

03 Advanced Photoshop

WEEKS 7-9 · 18 H

TOPICS

- → Photo compositing and perspective
- → Frequency separation and professional retouching
- → Colour grading and mood
- → Advertising layouts
- → Mockups and product presentation
- → Non-destructive workflow with smart objects

HANDS-ON LAB

Create a campaign key visual and three retouched product images.

YOU FINISH WITH

A campaign key visual and retouched images.

04 Layout with InDesign

WEEKS 10-12 · 18 H

TOPICS

- → Grids, baselines and columns
- → Master pages, styles and long documents
- → Brochures, catalogues and magazines
- → Image handling and links
- → Preflight and packaging files
- → Interactive and export-ready PDFs

HANDS-ON LAB

Lay out a 16-page brochure or catalogue with a real grid and styles.

YOU FINISH WITH

A 16-page multi-page document.

05 Packaging design

WEEKS 13-14 · 12 H

TOPICS

- → Structure and dielines
- → Labels, boxes and bags
- → Materials, finishes and foils
- → Barcodes and mandatory information
- → Mockups and presenting packaging

HANDS-ON LAB

Design packaging for a product from a supplied dieline.

YOU FINISH WITH

A packaging design with dieline and mockup.

06 Visual design in Figma

WEEKS 15-16 · 12 H

TOPICS

- → Frames, auto layout and components
- → Design systems and styles
- → Social and web campaign visuals
- → Handing designs to developers
- → Collaborating and giving feedback

HANDS-ON LAB

Build a small design system and a landing-page visual for the brand.

YOU FINISH WITH

A component library and a landing-page design.

07 Print production and prepress

WEEK 17 · 6 H

TOPICS

- → Colour management and proofing
- → Spot colours and special inks
- → Trapping, overprint and bleed
- → Working with a press and checking proofs
- → Common production mistakes

HANDS-ON LAB

Prepare final files for a press and review a proof.

YOU FINISH WITH

Production-ready files.

08 Client workflow and pricing

WEEK 18 · 6 H

TOPICS

- → Briefs, scope and revision rounds
- → Proposals, quotes and contracts
- → Pricing your time and your value
- → Presenting concepts and handling feedback
- → Handing over and archiving files

HANDS-ON LAB

Write a proposal and a pricing sheet, then role-play a feedback meeting.

YOU FINISH WITH

A proposal, a pricing sheet and a handover checklist.

09 Capstone brand project

WEEKS 19-20 · 12 H

TOPICS

- → Choose a real brand brief
- → Develop identity, applications and packaging
- → Assemble a case study
- → Present to a review panel

HANDS-ON LAB

Deliver a complete brand project as a studio would.

YOU FINISH WITH

A complete brand project and portfolio case study.

CAPSTONE PROJECT

A complete brand project: identity and guidelines, packaging, a printed brochure or catalogue, and campaign visuals, presented as a portfolio case study.

WHERE IT LEADS

- ✓ Graphic designer
- ✓ Brand designer
- ✓ Packaging designer
- ✓ Art director (junior)
- ✓ Studio or freelance designer

INTERNSHIP ROLE

Brand design intern. Work alongside the team on real tasks, with feedback.

GOOD NEXT STEPS

Video Editing | Digital Marketing

COURSE 8 OF 9

ADVANCED

Backend Development

16	3 × 2 h	96 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

The server side of real systems. You build REST APIs with Node.js, design databases in PostgreSQL, add secure logins, handle files, email and payments, test your work, and deploy and monitor it.

WHO IT IS FOR

People who can already write basic JavaScript and want to build what runs behind websites and apps: orders, users, payments and reports.

BEFORE YOU START

Basic JavaScript. If you have finished our Web Development course, you are ready.

BY THE END YOU CAN

- ✓ Design and build a REST API with validation and error handling.
- ✓ Model data and write queries in PostgreSQL.
- ✓ Implement authentication, roles and common security protections.
- ✓ Write tests and deploy a service with logs.
- ✓ Explain trade-offs in caching, performance and structure.

TOOLS: Node.js | Express | PostgreSQL | An ORM (Drizzle or Prisma) | Postman | Git | Docker (intro)

Plan at a glance

MODULE	WHEN	HOURS
01 Node.js and server JavaScript	Weeks 1–2	12 h
02 HTTP and REST APIs	Weeks 3–4	12 h
03 SQL and PostgreSQL	Weeks 5–7	18 h
04 ORMs and data access	Week 8	6 h
05 Authentication and security	Weeks 9–10	12 h
06 Files, email and payments	Week 11	6 h
07 Testing	Week 12	6 h
08 Caching and performance	Week 13	6 h
09 Deployment and operations	Week 14	6 h
10 Capstone project	Weeks 15–16	12 h
Total	**16 weeks**	**96 h**

Module by module

01 Node.js and server JavaScript

WEEKS 1–2 · 12 H

TOPICS

- → Running JavaScript outside the browser
- → Modules, packages and npm
- → The event loop, async code and promises
- → Working with the file system
- → Environment variables and configuration
- → TypeScript basics for backend code

HANDS-ON LAB

Write command-line tools that read, transform and write files.

YOU FINISH WITH

Two working command-line tools.

02 HTTP and REST APIs

WEEKS 3-4 · 12 H

TOPICS

- → HTTP methods, status codes and headers
- → Designing resources and endpoints
- → Express routing and middleware
- → Validating input with schemas
- → Error handling and consistent responses
- → Pagination, filtering and versioning
- → API documentation

HANDS-ON LAB

Build a complete CRUD API and document it.

YOU FINISH WITH

A documented CRUD API.

03 SQL and PostgreSQL

WEEKS 5-7 · 18 H

TOPICS

- → Tables, keys and relationships
- → Normalisation, and when to bend it
- → SELECT, JOIN, GROUP BY and subqueries
- → Indexes and reading query plans
- → Transactions and constraints
- → Migrations
- → Designing a schema from a business description

HANDS-ON LAB

Design a schema for an orders and stock system, then write the reports it needs.

YOU FINISH WITH

A normalised schema, seed data and a set of report queries.

04 ORMs and data access

WEEK 8 · 6 H

TOPICS

- → Why use an ORM, and its limits
- → Schema, relations and queries
- → Migrations in a team
- → Raw SQL when you need it

HANDS-ON LAB

Move your API from raw queries to an ORM.

YOU FINISH WITH

An API backed by an ORM with migrations.

05 Authentication and security

WEEKS 9-10 · 12 H

TOPICS

- → Password hashing and storage
- → Sessions and tokens
- → Roles and permissions
- → Input validation and injection attacks
- → CORS, CSRF and rate limiting
- → Secrets and safe configuration
- → A security checklist for your API

HANDS-ON LAB

Add registration, login, roles and rate limiting to your API and attack it yourself.

YOU FINISH WITH

A secured API with roles and a security checklist.

06 Files, email and payments

WEEK 11 · 6 H

TOPICS

- → File uploads and storage
- → Sending email reliably
- → Webhooks
- → How payment gateways work, safely
- → Background jobs and queues (intro)

HANDS-ON LAB

Add file uploads, email notifications and a webhook to your API.

YOU FINISH WITH

An API with uploads, email and a webhook.

07 Testing

WEEK 12 · 6 H

TOPICS

- → Unit and integration tests
- → Testing endpoints and databases
- → Test data and isolation
- → What to test, and what not to

HANDS-ON LAB

Write tests that cover the critical paths of your API.

YOU FINISH WITH

A test suite for the API.

08 Caching and performance

WEEK 13 · 6 H

TOPICS

- → Where time goes in a request
- → Caching strategies and Redis basics
- → Query optimisation
- → Pagination and large data sets

HANDS-ON LAB

Profile a slow endpoint, then make it faster and prove it.

YOU FINISH WITH

A before-and-after performance report.

09 Deployment and operations

WEEK 14 · 6 H

TOPICS

- → Hosting options and environments
- → Docker basics
- → CI: running tests on every push
- → Logging, monitoring and error tracking
- → Backups and recovery

HANDS-ON LAB

Deploy the API with automatic tests and logging.

YOU FINISH WITH

A deployed API with a CI pipeline and logs.

10 Capstone project

WEEKS 15–16 · 12 H

TOPICS

- → Choose a real business system
- → Design the schema and API
- → Build, test and secure it
- → Deploy and present

HANDS-ON LAB

Build a complete API for a business such as orders, stock or installments.

YOU FINISH WITH

A deployed, tested and documented business API.

CAPSTONE PROJECT

A production-style API for a real business system, with a PostgreSQL schema, authentication and roles, tests, a CI pipeline, and a deployed, documented service.

WHERE IT LEADS

- ✓ Backend developer (junior)
- ✓ API developer
- ✓ Database developer
- ✓ Freelance backend developer

INTERNSHIP ROLE

Backend intern. Work alongside the team on real tasks, with feedback.

GOOD NEXT STEPS

Full Stack Development

COURSE 9 OF 9

ADVANCED

Full Stack Development

32	4 × 2 h	256 h	4–8 wk
WEEKS	PER WEEK	CLASS TIME	INTERNSHIP

Estimated durations. Class time is hands-on sessions only, not self-practice or the internship.

The complete developer path, from zero to a deployed application. You learn the web foundations, JavaScript and TypeScript, React and Next.js, then Node.js, PostgreSQL, authentication and deployment, and finish by building a full business application.

WHO IT IS FOR

Beginners who want one complete path to becoming a developer, and learners who want to build whole products as a freelancer or inside a company.

BEFORE YOU START

None. This is the longest path and it starts from zero.

BY THE END YOU CAN

- ✓ Build a complete web application from interface to database.
- ✓ Use React and Next.js the way professional teams do.
- ✓ Design databases and secure APIs.
- ✓ Write tests and ship through a repeatable deployment process.
- ✓ Plan, build and present a product as a small team would.

TOOLS HTML and CSS · JavaScript and TypeScript · React · Next.js · Node.js · PostgreSQL · Tailwind CSS · Git and GitHub

Plan at a glance

MODULE	WHEN	HOURS
01 Web foundations	Weeks 1–4	32 h
02 JavaScript	Weeks 5–8	32 h
03 TypeScript	Weeks 9–10	16 h
04 Git, GitHub and Tailwind	Week 11	8 h
05 React	Weeks 12–16	40 h
06 Next.js	Weeks 17–20	32 h
07 Node.js and APIs	Weeks 21–23	24 h
08 PostgreSQL and ORMs	Weeks 24–26	24 h
09 Authentication and security	Weeks 27–28	16 h
10 Testing and deployment	Week 29	8 h
11 Capstone project	Weeks 30–32	24 h
Total	**32 weeks**	**256 h**

Module by module

01 Web foundations

WEEKS 1–4 · 32 H

TOPICS

- How the web works
- Semantic HTML and forms
- CSS: box model, Flexbox and Grid
- Responsive design
- Accessibility basics
- Building pages from designs

HANDS-ON LAB

Build and publish three pages from design files.

YOU FINISH WITH

A published responsive multi-page site.

02 JavaScript

WEEKS 5-8 · 32 H

TOPICS

- → Variables, functions and scope
- → Arrays, objects and array methods
- → The DOM and events
- → Fetch, promises and async/await
- → Error handling and debugging
- → Modules and tooling

HANDS-ON LAB

Solve problem sets and build three browser apps, one using a public API.

YOU FINISH WITH

Three working JavaScript apps.

03 TypeScript

WEEKS 9-10 · 16 H

TOPICS

- → Types, interfaces and generics
- → Typing functions and data
- → Narrowing and unions
- → Using TypeScript in a real project

HANDS-ON LAB

Convert a JavaScript app to TypeScript.

YOU FINISH WITH

A fully typed app.

04 Git, GitHub and Tailwind

WEEK 11 · 8 H

TOPICS

- → Commits, branches and pull requests
- → Working in a team repository
- → Tailwind: utilities, variants and components

HANDS-ON LAB

Build a team repository with reviewed pull requests.

YOU FINISH WITH

A repository with a clean history, styled with Tailwind.

05 React

WEEKS 12-16 · 40 H

TOPICS

- → Components, props and composition
- → State and effects
- → Hooks and custom hooks
- → Forms and validation
- → Data fetching and loading states
- → Routing and context
- → Structuring a larger app

HANDS-ON LAB

Build a data-driven React application with forms and routing.

YOU FINISH WITH

A multi-page React application.

06 Next.js

WEEKS 17-20 · 32 H

TOPICS

- → Routing, layouts and pages
- → Server and client components
- → Data fetching and caching
- → Route handlers and forms
- → Metadata, images and performance
- → Deployment

HANDS-ON LAB

Rebuild your React application in Next.js with server-side data.

YOU FINISH WITH

A deployed Next.js application.

07 Node.js and APIs

WEEKS 21-23 · 24 H

TOPICS

- → Node and Express
- → REST design, validation and errors
- → Middleware
- → Calling third-party APIs
- → Documenting an API

HANDS-ON LAB

Build an API that your frontend uses.

YOU FINISH WITH

A documented REST API connected to the frontend.

08 PostgreSQL and ORMs

WEEKS 24-26 · 24 H

TOPICS

- → Tables, relations and keys
- → Queries, joins and indexes
- → Migrations and seed data
- → Using an ORM
- → Designing a schema from a business brief

HANDS-ON LAB

Design the schema for your capstone and load it with realistic data.

YOU FINISH WITH

A designed and seeded database.

09 Authentication and security

WEEKS 27-28 · 16 H

TOPICS

- → Registration, login and sessions
- → Roles and permissions
- → Protecting against common attacks
- → Handling secrets
- → A security review

HANDS-ON LAB

Add roles and protected routes to your application.

YOU FINISH WITH

A secured application with roles.

10 Testing and deployment

WEEK 29 · 8 H

TOPICS

- → Unit and end-to-end tests
- → Environment variables and configuration
- → CI and deployment
- → Monitoring and fixing production issues

HANDS-ON LAB

Add tests and a deployment pipeline.

YOU FINISH WITH

A tested, continuously deployed application.

11 Capstone project

WEEKS 30–32 · 24 H

TOPICS

- → Choose and scope a real business application
- → Design the data, API and screens
- → Build in weekly milestones
- → Test, deploy and present

HANDS-ON LAB

Build a business application, such as a small ERP, as a team.

YOU FINISH WITH

A deployed full application, with an admin area, and a final presentation.

CAPSTONE PROJECT

A deployed full stack business application, for example a small ERP, with logins and roles, a PostgreSQL database, an admin area, tests, and a final presentation.

WHERE IT LEADS

- ✓ Full stack developer (junior)
- ✓ Frontend or backend developer
- ✓ Freelance full stack developer
- ✓ Startup product developer

INTERNSHIP ROLE

Full stack intern. Work alongside the team on real tasks, with feedback.

NEXT

This is the end of the path.

AL-KAUSAR
COMPUTER INSTITUTE

JOIN THE ACADEMY

Ready to start? *Apply.*

Send the admission form on our website or message us on WhatsApp. We reply with the fee, the next batch dates and what to bring, in writing, before you decide.

WEB	kataf.pk/apply
EMAIL	hello@kataf.pk
VISIT	Islamabad, Pakistan · Mon–Sat, 10:00–19:00
VERIFY	kataf.pk/verify